



“When GPS fails and RF links jam, SUTRA geonavigates and locates survivors in real-time.”

PROJECT SUTRA.

Swarm Unified Tactical Reconnaissance Architecture — decentralized multi-UAV flight, Deep JSCC neural zero-cliff video, and 3.59cm terrain-corrected DEM survivor geolocation.

01
DECENTRALIZED GNC

50Hz OFFBOARD RATE

PX4 MicroXRCE + VIO EKF2 Odometry

02
SEMANTIC WIRELESS

-5dB JAM RESILIENCE

Deep JSCC Neural Zero-Cliff Video

03
TRI-MODAL & 3D GEO

3.59cm RAYCAST
PRECISION

WGS84 3D DEM Slope Correction

CORE ARCHITECTURE TEAM (OFFGRID)

Nikhil

Tech Lead · Subsys A & B

GNC & COMMS

Vedanth Sai Ram

Lead · Subsystem C

AI PERCEPTION

Siva Kesava

Lead · Subsystem D

3D GIS GCS

Harika

Operations Lead · Subsys E

AUDIT & OPS

RESEARCH-BACKED & EMPIRICALLY VALIDATED

Rohith Kumar

Lead · Subsystem F

FIELD CONOPS